

Lab 13 – Introduction with Tkinter

Programming Exercise:

1. Write a program which will make a scientific calculator.
2. Give proper color to your calculator.

Source Code:

```
import tkinter as tk
import math # Importing math module for scientific functions like sin, cos, tan, sqrt, etc.

# Define the function that handles button clicks
def click(event):
    """
    This function handles the button click event.
    It checks which button is clicked and updates the entry widget accordingly.
    """
    # Get the text of the button that was clicked (such as '7', 'C', 'sin', '+=', etc.)
    button_text = event.widget.cget("text")

    # Get the current input from the entry widget
    current_input = entry.get()

    # Logic for "=" button (Evaluate expression)
    if button_text == "=":
        try:
            # Use eval() to evaluate the mathematical expression in the entry widget
            result = str(eval(current_input))
            entry.delete(0, tk.END) # Clear the entry widget before displaying the result
            entry.insert(0, result) # Display the result in the entry widget
        except Exception as e: # If there's an error, like invalid input, display "Error"
            entry.delete(0, tk.END)
            entry.insert(0, "Error")
```

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# Logic for "C" button (Clear entry)
elif button_text == "C":
    entry.delete(0, tk.END) # Clear all content in the entry widget

# Logic for square root button (√)
elif button_text == "√":
    try:
        # The square root function requires a non-negative number. We use math.sqrt() to calculate
        the square root
        result = str(math.sqrt(float(current_input))) # Convert input to float and calculate square
        root
        entry.delete(0, tk.END)
        entry.insert(0, result)
    except Exception as e: # If the input is invalid (like negative number), display "Error"
        entry.delete(0, tk.END)
        entry.insert(0, "Error")

# Logic for square button (x²)
elif button_text == "x²":
    try:
        # To calculate square of the number, we raise it to the power of 2
        result = str(float(current_input) ** 2) # Convert input to float and calculate square
        entry.delete(0, tk.END)
        entry.insert(0, result)
    except Exception as e: # If the input is invalid (like empty), display "Error"
        entry.delete(0, tk.END)
        entry.insert(0, "Error")

# Logic for trigonometric functions (sin, cos, tan)
elif button_text == "sin":
    try:
        # The input is assumed to be in degrees, so we convert it to radians for the math.sin()
        function
        result = str(math.sin(math.radians(float(current_input)))) # Convert degrees to radians and
        calculate sin
        entry.delete(0, tk.END)
        entry.insert(0, result)
    except Exception as e: # If input is invalid, show "Error"
        entry.delete(0, tk.END)
        entry.insert(0, "Error")

elif button_text == "cos":
    try:
        # Similar to sin, for cos, convert input to radians and calculate the cosine value
        result = str(math.cos(math.radians(float(current_input))))
        entry.delete(0, tk.END)
        entry.insert(0, result)
    except Exception as e:
        entry.delete(0, tk.END)
        entry.insert(0, "Error")

elif button_text == "tan":
    try:
        # Similar to sin, for tan, convert input to radians and calculate the tangent value
        result = str(math.tan(math.radians(float(current_input))))
        entry.delete(0, tk.END)
        entry.insert(0, result)
    except Exception as e:
        entry.delete(0, tk.END)
        entry.insert(0, "Error")

# Logic for decimal point (".") button
elif button_text == ".":
    if "." not in current_input.split()[-1]: # Check if there is already a decimal point in the last
    number
        entry.insert(tk.END, button_text) # If not, add the decimal point

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        entry.insert(tk.END, button_text) # If not, add the decimal point
    else:
        pass # If there's already a decimal point, do nothing

    # If any other button is clicked (numbers or operators), add the corresponding text to the entry
    widget
    else:
        entry.insert(tk.END, button_text) # Insert button's text at the end of current input

# Create the main window for the calculator
root = tk.Tk()
root.geometry("400x600") # Set the window size (400px width, 600px height)
root.title("Scientific Calculator") # Set the window title
root.config(bg="#fce4ec") # Set the background color of the window (light pink)

# Create the entry widget (where input and output will be displayed)
entry = tk.Entry(root, font=("Helvetica", 20), justify="right", width=15, bg="#f8bbd0", fg="black",
relief="solid")
entry.grid(row=0, column=0, columnspan=4, padx=20, pady=20) # Place the entry widget on the grid

# Define the layout of the buttons in a list (each button has its text and grid position)
buttons = [
    ("7", 1, 0), ("8", 1, 1), ("9", 1, 2), ("/", 1, 3), # First row (numbers and division)
    ("4", 2, 0), ("5", 2, 1), ("6", 2, 2), ("*", 2, 3), # Second row (numbers and multiplication)
    ("1", 3, 0), ("2", 3, 1), ("3", 3, 2), ("-", 3, 3), # Third row (numbers and subtraction)
    ("C", 4, 0), ("0", 4, 1), ("=", 4, 2), ("+", 4, 3), # Fourth row (clear, zero, equals, addition)
    ("√", 5, 0), ("x²", 5, 1), # Fifth row (square root and square)
    ("sin", 6, 0), ("cos", 6, 1), ("tan", 6, 2), # Sixth row (sin, cos, tan)
    (".", 7, 0) # Add the decimal point (.) button in the next row
]

# Create the buttons and place them on the grid
for (text, row, col) in buttons:
    # Create a button for each entry in the buttons list
    button = tk.Button(root, text=text, font=("Helvetica", 18), width=5, height=2, relief="raised",
bg="#ff66b2",
fg="white")
    button.grid(row=row, column=col, padx=10, pady=10) # Place each button on the grid

    # Bind the button click event to the click() function
    button.bind("<Button-1>", click) # Bind each button's click event to the function

# Start the Tkinter event loop to run the calculator
root.mainloop() # Keeps the window running until closed by the user

```

Output:



3. Design any creative application from the exercises that have been taught to you in this lab. Such as POS systems for any Pharmacy, or any grocery store, or BBQ restaurant, or Pizza Hut or any application which you like.

Source Code:

```
import tkinter as tk
from tkinter import messagebox

# Create the main window
root = tk.Tk()
root.title("POS System")
root.geometry("600x400")
root.config(bg="#FFB6C1") # Set background color of the main window (blue theme)

# Function to add product to cart
def add_to_cart(product_name, price):
    if product_name in cart:
        cart[product_name]["quantity"] += 1
    else:
        cart[product_name] = {"price": price, "quantity": 1}

    update_cart_display()

# Function to update the cart display
def update_cart_display():
    cart_display.delete(1.0, tk.END)
    total = 0
    for product, details in cart.items():
        cart_display.insert(tk.END, f"{product} x {details['quantity']} = ${details['price']} * {details['quantity']}\n")
        total += details['price'] * details['quantity']

    cart_display.insert(tk.END, f"\nTotal: ${total:.2f}")

# Function to checkout and process payment
def checkout():
    total = 0
    for details in cart.values():
        total += details["price"] * details["quantity"]

    try:
        payment = float(payment_entry.get())
        if payment < total:
            messagebox.showerror("Error", "Not enough funds")
        else:
            change = payment - total
            messagebox.showinfo("Success", f"Payment successful! Change: ${change:.2f}")
            generate_receipt()
    except ValueError:
        messagebox.showerror("Error", "Please enter a valid payment amount")

# Function to generate receipt
def generate_receipt():
    receipt_window = tk.Toplevel()
    receipt_window.title("Receipt")
    receipt_window.geometry("300x300")
    receipt_window.config(bg="#DB7093")

    receipt_label = tk.Label(receipt_window, text="Receipt", font=("Arial", 16), fg="black",
                             bg="#FFB6C1")
    receipt_label.pack(pady=10)
```



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        receipt_text = tk.Text(receipt_window, height=10, width=30, font=("Arial", 12), bg="#f8f9fa",
fg="black")
        receipt_text.pack()

        total = 0
        for product, details in cart.items():
            receipt_text.insert(tk.END, f"{product} x {details['quantity']} = ${details['price']} *
details['quantity']}\n")
            total += details['price'] * details['quantity']

        receipt_text.insert(tk.END, f"\nTotal: ${total:.2f}\n")
        receipt_text.insert(tk.END, f"Payment: ${payment_entry.get()}\n")
        receipt_text.insert(tk.END, f"Change: ${float(payment_entry.get()) - total:.2f}")

# Initialize cart (dictionary)
cart = {}

# Create a Frame for the product list
product_frame = tk.Frame(root, bg="#FF69B4", bd=5)
product_frame.pack(pady=10, padx=10, fill=tk.X)

# Add product buttons (8 items)
products = [
    ("Aspirin", 5.00),
    ("Cough Syrup", 8.00),
    ("Band-Aid", 2.00),
    ("Paracetamol", 4.00),
    ("Antiseptic Cream", 6.00),
    ("Vitamins", 10.00),
    ("Cold Pack", 3.00),
    ("Thermometer", 12.00)
]

# Create product buttons dynamically
for product, price in products:
    button = tk.Button(product_frame, text=f"{product} - ${price:.2f}", font=("Arial", 12), bg="#FFB6C1",
fg="black",
                        command=lambda product=product, price=price: add_to_cart(product, price))
    button.pack(side=tk.LEFT, padx=5)

# Create a Frame for the cart
cart_frame = tk.Frame(root, bg="#FFB6C1", bd=5)
cart_frame.pack(pady=10, padx=10, fill=tk.X)

# Display cart contents
cart_display = tk.Text(cart_frame, height=10, width=40, font=("Arial", 12), bg="#f8f9fa", fg="black")
cart_display.pack()

# Create a Frame for payment and checkout
checkout_frame = tk.Frame(root, bg="#FF69B4", bd=5)
checkout_frame.pack(pady=10, padx=10, fill=tk.X)

# Entry for payment amount
payment_label = tk.Label(checkout_frame, text="Enter Payment Amount:", font=("Arial", 12), fg="black",
bg="#FFB6C1")
payment_label.pack(side=tk.LEFT, padx=5)

payment_entry = tk.Entry(checkout_frame, font=("Arial", 12), bg="fff", fg="black")
payment_entry.pack(side=tk.LEFT, padx=5)

# Checkout button
checkout_button = tk.Button(checkout_frame, text="Checkout", font=("Arial", 12), bg="#007bff",
fg="white",
                        command=checkout)
checkout_button.pack(side=tk.LEFT, padx=5)

# Run the main loop
root.mainloop()

```

Output:

POS System

Aspirin - \$5.00	Cough Syrup - \$8.00	Band-Aid - \$2.00	Paracetamol - \$4.00	Antiseptic Cream - \$6.00	Vitamins - \$10.00	Cold Pack - \$3.00	Thermometer - \$12.00
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Enter Payment Amount:

POS System

Aspirin - \$5.00	Cough Syrup - \$8.00	Band-Aid - \$2.00	Paracetamol - \$4.00	Antiseptic Cream - \$6.00	Vitamins - \$10.00	Cold Pack - \$3.00	Thermometer - \$12.00
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Thermometer x 1 = \$12.0
Cold Pack x 3 = \$9.0
Paracetamol x 3 = \$12.0
Band-Aid x 8 = \$16.0
Cough Syrup x 1 = \$8.0
Aspirin x 2 = \$10.0
Antiseptic Cream x 3 = \$18.0
Vitamins x 3 = \$30.0
Total: \$115.00

Enter Payment Amount: 250.80

POS System

Aspirin - \$5.00	Cough Syrup - \$8.00	Band-Aid - \$2.00	Paracetamol - \$4.00	Antiseptic Cream - \$6.00	Vitamins - \$10.00	Cold Pack - \$3.00	Thermometer - \$12.00
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Thermometer x 1 = \$12.0
Cold Pack x 3 = \$9.0
Paracetamol x 3 = \$12.0
Band-Aid x 8 = \$16.0
Cough Syrup x 1 = \$8.0
Aspirin x 2 = \$10.0
Antiseptic Cream x 3 = \$18.0
Vitamins x 3 = \$30.0
Total: \$115.00

Enter Payment Amount: 250.80

Success
Payment successful! Change: \$135.80

POS System

Aspirin - \$5.00

Cough Syrup - \$8.00

Band-Aid - \$2.00

Paracetamol - \$4.00

Antiseptic Cream - \$6.00

Vitamins - \$10.00

Cold Pack - \$3.00

Thermometer - \$12.00

Thermometer x 1 = \$12.0
Cold Pack x 3 = \$9.0
Paracetamol x 3 = \$12.0
Band-Aid x 8 = \$16.0
Cough Syrup x 1 = \$8.0
Aspirin x 2 = \$10.0
Antiseptic Cream x 3 = \$18.0
Vitamins x 3 = \$30.0
Total: \$115.00

Enter Payment Amount: 250.80

Checkout

Receipt

Receipt

Thermometer x 1 = \$12.0
Cold Pack x 3 = \$9.0
Paracetamol x 3 = \$12.0
Band-Aid x 8 = \$16.0
Cough Syrup x 1 = \$8.0
Aspirin x 2 = \$10.0
Antiseptic Cream x 3 = \$18.0
Vitamins x 3 = \$30.0
Total: \$115.00